

Name: _____



Extending thinking about sequences

Task A: Concrete versus abstract sequences

1) This table shows us the history of men's 100 m sprint world record times.

Year	1940	1960	1980	2000	2020	2040	2060	2080	2100
Time (s)	10.22	10.06	9.9	9.74	9.58				

a) Identify a pattern and use it to predict the times up to the year 2100.

b) This is a concrete sequence. What is its physical limit?

Task B: Extending sequences in both directions

1) Scientists start studying the Penguin population on an island in the Southern Ocean annually and they find these numbers.

Year	1	2	3	4		
Population (000's)	160	200	250	312.5		

The population data forms a geometric sequence.

a) Move forwards to estimate the population next year and in two years time.

b) Move backwards to find how many years ago the population was under one hundred thousand.